

README - Replication Package

Firm Balance Sheet Liquidity, Monetary Policy Shocks, and Investment Dynamics

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1 Overview

This repository contains the data and code required to replicate all tables and figures in the manuscript “*Firm Balance Sheet Liquidity, Monetary Policy Shocks, and Investment Dynamics*.” The code conducts the full empirical analysis, including the cleaning, construction of relevant variables, and regression analysis in the paper employing data mostly from Compustat (S&P Global Market Intelligence, 2019), using Python, R, Julia and MATLAB. After conducting the core empirical analysis, the code runs the analysis of the computational structural general equilibrium model using Julia.

There is one main file (`Master_main.py`) that executes all of the required codes to generate the output of the replication package. This output contains 37 figures (with 69 subfigures altogether) and 12 tables in the paper, of which 10 are generated automatically by the code and 2 appendix tables (Tables B.1 and B.5) are compiled manually from partial `.txt` outputs generated by the code. The package also produces a number of auxiliary `.txt` files, including files reporting precise numbers cited in the manuscript that are not exactly legible from the figures or tables. The replicator should expect the code to run for about 22 hours.

All scripts rely exclusively on relative paths and presume the directory structure shown in Section 8 below.

2 Data Availability and Provenance Statements

2.1 Statement about Rights

- ✓ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- ✓ I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package.

2.2 Summary of Availability

- ✓ Some data cannot be made publicly available.

This project uses proprietary firm-level data from Compustat, CRSP, and Worldscope, accessed through institutional subscriptions. These raw and processed proprietary datasets are therefore not redistributed as part of the public replication package. Publicly available macroeconomic and financial series used in the analysis are included in the package where redistribution is permitted.

For the purpose of precise reproducibility checks, the non-public data used in the project have been made available to the JPE Data Editor under the Journal's procedures.

2.3 License for Data

The public replication package does not redistribute proprietary firm-level data from Compustat, CRSP, or Worldscope. These data remain subject to the licensing and access conditions imposed by their respective providers and by Wharton Research Data Services (WRDS), through which the data were accessed.

Publicly available data included in the replication package remain subject to the original terms of use and licensing conditions specified by their respective providers. The author does not claim authority to alter those terms.

2.4 Details on Each Data Source

PUBLIC **Federal Reserve Flow of Funds Accounts / Financial Accounts of the United States (FRB Z.1).** Contains annual balance sheet series for the U.S. nonfinancial corporate sector, among other sectors, from the Federal Reserve's Flow of Funds Accounts (Board of Governors of the Federal Reserve System, 2016), used to construct measures of the composition of firms' liquid asset holdings. The file was accessed and downloaded on October 3, 2016 using the Data Download Program on the website of the Federal Reserve, at <https://www.federalreserve.gov/datadownload/default.htm>. The data are provided as part of this archive.

Datafile: `data/raw/FRB_Z1.csv`.

PUBLIC **3-month Treasury Bill rate.** Contains the annual series, constructed by averaging, of the 3-month Treasury Bill Secondary Market Rate, Discount Basis (Board of Governors of the Federal Reserve System (US), 2026). The series was first accessed via FRED on April 5, 2024, and the version archived in this replication package was downloaded on May 11, 2026. The data are provided as part of this archive.

Datafile: `data/raw/TB3MS.csv`.

PUBLIC 1-year Treasury constant maturity rate. Contains the daily series of the 1-year Treasury constant maturity rate (Board of Governors of the Federal Reserve System (US), 2020). The series was accessed and downloaded via FRED on August 12, 2020. The data are provided as part of this archive.

Datafile: `data/raw/DGS1.csv`.

PUBLIC Gross Domestic Product: Implicit Price Deflator. Contains the quarterly series of the Gross Domestic Product implicit price deflator (U.S. Bureau of Economic Analysis, 2020). The series was accessed and downloaded via FRED on August 12, 2020. The data are provided as part of this archive.

Datafile: `data/raw/GDPDEF_Q.csv`.

PUBLIC Effective Federal Funds Rate. Contains the quarterly series, constructed by averaging, of the Effective Federal Funds Rate (Board of Governors of the Federal Reserve System (US), 2017). The series was accessed and downloaded via FRED on March 31, 2017. The data are provided as part of this archive.

Datafile: `data/raw/FEDFUNDS_Q.xls`.

PUBLIC Real Gross Domestic Product, Chained Dollars. Contains the quarterly series of real Gross Domestic Product from Table 1.1.6 of the National Income and Product Accounts of the U.S. Bureau of Economic Analysis (U.S. Bureau of Economic Analysis, 2017a). The series was accessed and downloaded on July 21, 2017. The data are provided as part of this archive.

Datafile: `data/raw/GDPr_Q_BEA.xls`.

PUBLIC Gross Value Added Deflator. Contains the quarterly gross value added price index series from Table 1.3.4, “Price Indexes for Gross Value Added by Industry,” of the U.S. Bureau of Economic Analysis (U.S. Bureau of Economic Analysis, 2017b). The series was accessed and downloaded on August 29, 2017. The data are provided as part of this archive.

Datafile: `data/raw/GVADEF_Q_BEA.xls`.

PUBLIC Business Dynamics Statistics Economy-wide and Firm Age datasets. Contain establishment and firm dynamics data from the U.S. Census Bureau’s Business Dynamics Statistics at the economy-wide level (U.S. Census Bureau, 2018a) and by firm age (U.S. Census Bureau, 2018b). The data are from the 2018 BDS release and were accessed and downloaded on the website of the U.S. Census on February 16 and 15, 2021, respectively. The data are provided as part of this archive.

Datafiles: `data/raw/bds2018.csv`, `data/raw/bds2018_fage.csv`.

PUBLIC Jarocinski and Karadi (2020a) monetary policy shocks dataset. Contains the monetary policy shock dataset used in the empirical analysis of Jarocinski and Karadi (2020a), obtained in late 2019 as the file `data_fig4.csv` from the replication package accompanying Jarocinski and Karadi (2020a) when the paper was forthcoming in AEJ:

Macroeconomics. The author has verified that the file used here is identical to that contained in the currently archived public replication package (Jarocinski and Karadi, 2020b). The data are provided as part of this archive.

Datafile: `data/raw/JK_data_fig4.csv`.

PUBLIC Romer-Romer monetary shocks series, reestimated by Ramey (2016b).

Contains the monthly monetary shock series, reestimated by Ramey (2016b) for the 1983m1–2007m12 period, based on updates to the Romer and Romer (2004) dataset by Wieland and Yang (2017). The data were originally obtained in September 2017 from the website of Valerie Ramey as part of the Data and Programs provided to accompany Ramey (2016b). The series was extracted as the variable `RRSHOCK83` from the file `Monetarydat.xlsx` in that package (Ramey, 2016a). The data are provided as part of this archive.

Datafile: `data/raw/sffr_RR.csv`.

PUBLIC U.S. time series data on real quantities for aggregate local projection

estimation. Contains the quarterly U.S. aggregate macroeconomic series used in the aggregate local projection workflow of Luetticke (2021b), updated for this replication package for the aggregate local projections reported in Appendix A.6 in the paper. More specifically, these data contain the aggregate series of real consumption (U.S. Bureau of Economic Analysis, 2025g), investment (U.S. Bureau of Economic Analysis, 2025f), and government spending (U.S. Bureau of Economic Analysis, 2025e), accessed via FRED on June 19, 2025. The data are provided as part of this archive.

Datafiles: `data/raw/LPagg_data/PCECC96.xlsx`, `data/raw/LPagg_data/GPDIC1.xlsx`, `data/raw/LPagg_data/GCEC1.xlsx`.

PUBLIC U.S. time series data on consumption and investment deflators for ag-

gregate local projection estimation. Contains the quarterly U.S. consumption and investment deflator series used in the aggregate local projections reported in Appendix A.6 in the paper. More specifically, these data contain the deflator series of nondurable consumption (U.S. Bureau of Economic Analysis, 2025d), gross private domestic investment (U.S. Bureau of Economic Analysis, 2025c), nonresidential structures (U.S. Bureau of Economic Analysis, 2025b), and nonresidential equipment (U.S. Bureau of Economic Analysis, 2025a) accessed via FRED on June 19, 2025. The data are provided as part of this archive.

Datafiles in `data/raw/LPagg_data/`: `DNDGRD3Q086SBEA.xlsx`, `A006RD3Q086SBEA.xlsx`, `A009RD3Q086SBEA.xlsx`, `Y033RD3Q086SBEA.xlsx`.

PROPRIETARY Compustat North America Quarterly.

Contains quarterly firm-level accounting and financial statement data from Compustat North America Quarterly (S&P Global Market Intelligence, 2019), between 1961–2018 used to construct the main panel of public-firm balance-sheet, cash holdings, debt, and capital accumulation variables for the empirical analysis. The dataset was accessed and downloaded via WRDS (Wharton Research Data Services, 2026) on March 29, 2019. The data are not provided as part of the public

replication package.

Datafile: `proprietary-data-not-for-publication/raw/CS_data_raw.csv`.

PROPRIETARY CRSP U.S. Stock Database. Contains daily firm-level equity market data between 1950–2019 from the CRSP U.S. Stock Database (Center for Research in Security Prices (CRSP), 2020), including prices, returns, shares outstanding, and market capitalization. These data are used to construct the distance-to-default measure used as a control in the empirical analysis. The dataset was accessed and downloaded via WRDS (Wharton Research Data Services, 2026) on July 31, 2020. The data are not provided as part of the public replication package.

Datafile: `proprietary-data-not-for-publication/raw/CRSP_data_raw.csv`.

PROPRIETARY CRSP/Compustat Merged (CCM) Database. Contains the CRSP/Compustat linking table from the CRSP/Compustat Merged (CCM) Database (Center for Research in Security Prices (CRSP), 2019), used to merge Compustat firm-level accounting data with CRSP market data in the empirical analysis. The dataset was accessed and downloaded via WRDS (Wharton Research Data Services, 2026) on July 30, 2019. The data are not provided as part of the public replication package.

Datafile: `proprietary-data-not-for-publication/raw/CCM_data_raw.csv`.

PROPRIETARY Firm Incorporation and Foundation Date data from Refinitiv Worldscope. Contains an Excel export from Worldscope (Refinitiv, 2019b), accessed through Datastream (Refinitiv, 2019a), with firm-level identifiers together with firms’ incorporation and foundation dates. More specifically, it includes the columns `Type`, `NAME`, `WC05601`, `WC06004`, `WC18272`, `WC18273`, and one row per firm, split across 24 Excel sheets. The incorporation dates are used to construct a measure of firm age that enters the empirical analysis as a control variable. The dataset was accessed through Datastream on April 9, 2019. The data are not provided as part of the public replication package.

Datafile: `proprietary-data-not-for-publication/raw/Wscope_data_raw.xlsx`.

2.5 Compliance with Public Data Format

As required by the JPE Data and Code Policy (Point 1.4, “Format”), all publicly provided data files referenced above that are used in the code in `.xls` or `.xlsx` format are also accompanied in the replication package by corresponding copies in `.csv` format (separated by semicolon delimiters), with analogous, self-explanatory filenames. These copies are not employed by the procedures in the package.

2.6 Instructions for Restricted Data Access

Researchers interested in replicating the results using the proprietary firm-level data must obtain independent authorization from the relevant data providers.

2.7 Dataset List

Note that throughout the replication package, work related to the cash-to-assets, i.e. *liquidity* ratio, uses the variable mnemonic `cheat_rat`. This name derives from the standard Compustat mnemonics `che` (*Cash and Short-Term Investments*; `cheq` in quarterly Compustat) and `at` (*Total Assets*; `atq` in quarterly Compustat).

Table 1: Dataset list for the replication package

Path / File	Notes
data/raw/	
FRB_Z1.csv	Contains the raw annual data from the Federal Reserve's Flow of Funds Accounts, including the balance sheet for the U.S. nonfinancial corporate sector.
TB3MS.csv	Contains the raw annual series of the 3-month Treasury Bill rate.
DGS1.csv	Contains the raw daily series of the 1-year Treasury constant maturity rate.
GDPDEF_Q.csv	Contains the raw quarterly series of the U.S. Gross Domestic Product implicit price deflator.
FEDFUNDS_Q.xls	Contains the raw quarterly series of the Effective Federal Funds Rate.
GDPr_Q_BEA.xls	Contains the raw quarterly data of the U.S. real Gross Domestic Product (BEA Table 1.1.6).
GVADEF_Q_BEA.xls	Contains the raw quarterly gross value added price index data from (BEA Table 1.3.4).
bds2018.csv	Contains the raw economy-wide Business Dynamics Statistics data from the U.S. Census Bureau.
bds2018_fage.csv	Contains the raw firm-age Business Dynamics Statistics data from the U.S. Census Bureau.
JK_data_fig4.csv	Contains the Jarocinski and Karadi (2020a) monetary policy shock dataset.
sffr_RR.csv	Contains the Romer and Romer (2004) monetary shock series, as reestimated by Ramey (2016b).
data/raw/LPagg_data/	
PCECC96.xlsx	Contains the raw quarterly series of U.S. real personal consumption expenditures.

Path / File	Notes
GPDIC1.xlsx	Contains the raw quarterly series of U.S. real gross private domestic investment.
GCEC1.xlsx	Contains the raw quarterly series of U.S. government consumption expenditures and gross investment.
DNDGRD3Q086SBEA.xlsx	Contains the raw quarterly series of the implicit price deflator for U.S. nondurable consumption.
A006RD3Q086SBEA.xlsx	Contains the raw quarterly series of the implicit price deflator for U.S. gross private domestic investment.
A009RD3Q086SBEA.xlsx	Contains the raw quarterly series of the implicit price deflator for U.S. private nonresidential fixed investment in structures.
Y033RD3Q086SBEA.xlsx	Contains the raw quarterly series of the implicit price deflator for U.S. private nonresidential fixed investment in equipment.
proprietary-data-not-for-publication/raw/	
CS_data_raw.csv	Contains the raw quarterly Compustat North America firm-level accounting and financial statement data.
CRSP_data_raw.csv	Contains the raw daily CRSP equity market data.
CCM_data_raw.csv	Contains the raw CRSP/Compustat linking table used to merge Compustat firm-level accounting data with CRSP market data.
Wscope_data_raw.xlsx	Contains the raw Worldscope firm-level incorporation and foundation date data.
proprietary-data-not-for-publication/processed/CRSP_split	
CRSP_data_clean_X.csv	Contains the first cleaned iteration of the daily CRSP data, after dropping unused columns from the raw dataset and splitting it into smaller files to reduce RAM usage, with $X=0, \dots, 9$. Generated by <code>CRSP_cleaning_code.py</code> .
CRSP_data_DD_int_X.csv	Contains the daily CRSP data, merged with the Compustat variables in <code>CS_data_clean.csv</code> (see below) relevant for the later computation of distance-to-default, with $X=0, \dots, 9$. Generated by <code>CStoCRSP_merge.py</code> .
CRSP_data_DD_int_forpar_X.csv	Contains the daily CRSP data, as merged with Compustat, split into 60 partitions of firms in preparation for computing distance-to-default using parallelization, with $X=0, \dots, 59$. Generated by <code>CRSP_par_prep.py</code> .

Path / File	Notes
CRSP_data_DD_int_done_X.csv	Contains the daily CRSP data, as merged with Compustat, together with the outputs from distance-to-default computation, with $X=0, \dots, 59$. Generated by <code>DD_work_par_localwrapper.py</code> calling instances of <code>DD_work_run.py</code> .
proprietary-data-not-for-publication/processed/	
CS_data_clean.csv	Contains the quarterly Compustat data after initial cleaning of the <code>CS_data_raw.csv</code> file. Generated by <code>CS_cleaning_code.py</code> .
Wscope_data_compiled.csv	Contains the Worldscope data on firms' incorporation and foundation dates after compiling the data in the <code>Wscope_data_raw.xlsx</code> file into a single <code>.csv</code> file. Generated by <code>Wscope_compile_code.py</code> .
Micro_reg_data_compiled.csv	Contains the main Compustat panel dataset on which the empirical analysis of Section 2 in the paper is conducted. It is the result of further cleaning of <code>CS_data_clean.csv</code> , variable construction, and merging with the distance-to-default data from <code>CRSP_data_DD_int_done_X.csv</code> and the incorporation-date data from <code>Wscope_data_compiled.csv</code> . Generated by <code>Micro_reg_prep_code.py</code> .
p_data_saved.rds	Interim version of <code>Micro_reg_data_compiled.csv</code> after additional variable construction and preparation for analysis in R. Used in the firm-level debt issuance and local projection analyses. Generated and used by <code>Micro_reg_code.r</code> .
interim_output/	
CS_agg_cheat.t.csv	Contains the quarterly series of the aggregate cash-to-assets ratio, and the quartiles of the cross-sectional liquidity ratio distribution for firms in the Compustat panel. Generated by <code>Micro_reg_code.r</code> .
emp_dissreg_vals_ind(X)_dep(Y).csv	Contain the point estimates and confidence intervals from the event-study regressions estimated on Compustat data, depicted in Section 2.3 and Appendix A.3, with X referring to one of the "event indicators" <code>diss_ind</code> , <code>ndltisqat_rat</code> , or <code>kinvat_rat</code> ; and Y referring to one of the outcome variables <code>ndltisqat_rat</code> , <code>kinvat_rat</code> , or <code>dcheat_rat</code> . Generated by <code>Micro_reg_code.r</code> .

Path / File	Notes
<code>mod_dissreg_vals_dep(Y).csv</code>	Contain the point estimates from the event-study regressions estimated on model-generated data, depicted in Appendix A.3, with Y referring to one of the outcome variables <code>ndltisqat_rat</code> , <code>kinvat_rat</code> , or <code>dcheat_rat</code> . Generated by <code>Micro_model_reg_code.r</code> .
<code>Fig4b_nums_groupspec.Dfreq.txt</code>	Contains the leverage-liquidity group-specific debt issuance frequencies in Compustat data, as depicted in Figure 4b. Generated by <code>Micro_reg_code.r</code> .
<code>Fig**_slevsbetas_vec.q_vals(Y)(X)_IV_FE.csv</code> , <code>Fig**_slevsbetas_vec.q_cis(Y)(X)_IV_FE.csv</code>	Contain the point estimates and confidence intervals, respectively, of empirically estimated local projection IV coefficients, with X referring to the conditioning variable <code>cheat_rat</code> or <code>cheat_rat_dm</code> , and Y referring to the outcome variables <code>lk_stock</code> , <code>xintqdt_rat</code> , or <code>dltisqat_rat</code> . These estimates are for later use in plotting alongside the corresponding model estimates in Figures 6, B2, B3a, B3b, referred to by <code>**</code> in the filename. Generated by <code>Micro_reg_code.r</code> .
<code>F0FA_agg_cheat_t.csv</code>	Contains the annual series of the aggregate cash-to-assets ratio computed from the Fed Flow of Funds Accounts data for the U.S. nonfinancial corporate sector. Generated by <code>F0FA_analysis_code.py</code> .
interim_output/model_interim_output	
<code>df_model_panel.csv</code>	Contains the panel data on real and financial variables for 20,000 “public” firms for 40 quarters in the steady state of the calibrated quantitative model in the paper. Generated by <code>Master_model_work.jl</code> .
<code>IRFs_base.csv</code>	Contains the impulse responses of the nominal monetary policy rate, aggregate output, investment, and the real capital price in the baseline model, as depicted in Figure 5. Generated by <code>Master_model_work.jl</code> .
<code>anl_regcfs_vals_tempcheat.**_pub.jld</code>	Contains the model-implied coefficients from the regressions estimating heterogeneous capital accumulation responses to a monetary shock. These coefficients are depicted in Figure 7b, in the three corresponding scenarios referred to by <code>**</code> in the filename, as <code>base</code> , <code>rbtaureal</code> , or <code>rspiq</code> . Generated by <code>Master_model_work.jl</code> .
interim_output/model_interim_output/ savedsolns	

Path / File	Notes
<code>fsoln_**.jld</code>	Files containing the structural model firm's steady state solutions, including firm policy functions over the firm's idiosyncratic state space. ** in the filename refers to one of the three alternative model steady state scenarios considered in the computations: <code>base</code> , <code>base_phiw0</code> , or <code>hr_cf</code> . Generated by <code>Master_model_work.jl</code> .
<code>irf_**.jld</code>	Files containing the full equilibrium solution paths for structural model monetary shock scenarios, including aggregate quantities, prices, and firm policy functions over the firm's idiosyncratic state space in each period of the shock scenario. There is a variety of model exercises to which ** can refer. For conventional general equilibrium (GE) responses, ** refers to either <code>base</code> , <code>base_phiw0</code> , <code>hr_cf</code> , or <code>hr_cf_basephimresp</code> . For partial equilibrium (PE) responses, in which certain price response paths are taken from an alternative scenario and markets are not cleared, the filename also includes the mnemonic <code>_noGE</code> . For decompositions of responses into the effects of specific transmission channels (see Section 4.3.3 in the paper), the filename adds the suffix <code>_decomp_C</code> , with <code>C</code> referring to one of the channels <code>rbtaureal</code> , <code>rmtaureal</code> , <code>Q</code> , <code>MU</code> , or <code>M</code> . For objects corresponding to the low capital adjustment cost parameterization with $\varphi = 1/8$ (Appendix B.11.2), the filenames also include the suffix <code>_phik1d8</code> . Generated by <code>Master_model_work.jl</code> .

2.8 Variable Metadata and Documentation

For the third-party source datasets, the replication package does not reproduce full variable-by-variable codebooks. Instead, the raw variables follow the naming conventions and metadata of the original data providers, including Compustat, CRSP, Worldscope, the Federal Reserve, the U.S. Census Bureau, the U.S. Bureau of Economic Analysis, and FRED. The relevant source datasets and their provenance are documented in Section 2.4.

For the constructed datasets generated within the replication workflow, key variables used in the analysis are documented in the codebook provided in Section 9. Additional documentation is provided through the dataset descriptions, program descriptions, implementation notes, and variable names used in the code.

3 Computational Requirements

3.1 Software Requirements

- ✓ The replication package contains one or more programs to install all dependencies and set up the necessary directory structure.

The replication package was last run with the following software and package versions:¹

- Python 3.9.5
 - pandas 1.3.1
 - numpy 1.24.4
 - scipy 1.8.1
 - openpyxl 3.1.5
 - xlrd 2.0.2
 - A `requirements.txt` file is provided in the root directory of the replication package. It can be used to install the Python packages used in the workflow at the versions listed above (see Section 5 below for instructions).
- R 4.1.0
 - plm 2.4-1
 - plyr 1.8.6
 - lmtest 0.9-38
 - stargazer 5.2.2
 - lfe 2.8-6
 - moments 0.14
 - xtable 1.8.4
 - fixest 0.10.1
 - An `installR.packages.r` file is provided in the root directory of the replication package. It can be used to install the R packages used in the workflow at the versions listed above (see Section 5 below for instructions).
- Julia 1.9.1
 - LaTeXStrings 1.3.1
 - Interpolations 0.15.1
 - Roots 2.2.1
 - QuantEcon 0.16.6
 - CompEcon 0.4.0
 - JLD 0.13.5
 - CSV 0.10.14
 - Distributions 0.25.45
 - DataFrames 1.5.0
 - GLM 1.9.0

¹Importantly, the package was executed on a Mac mini with Intel architecture. Appendix A below provides additional notes for Apple Silicon Mac users (M1/M2/M3), which might be of help.

- `Binscatters` 0.4.0
- `Plots` 1.40.8
- `AverageShiftedHistograms` 0.8.9
- An `install_Julia_packages.jl` file is provided in the root directory of the replication package. It can be used to install the Julia packages used in the workflow at the versions listed above (see Section 5 below for instructions).
- MATLAB, version 9.12.0.1956245 (R2022a) Update 2
 - Statistics and Machine Learning Toolbox, version 12.3
- The code was last run on macOS Sequoia 15.4.1 (24E263).

3.2 Controlled Randomness

Random seeds are set repeatedly in the structural model code whenever random number generation is used. This is done to preserve reproducibility not only for full end-to-end execution, but also for modular execution of individual code blocks or subroutines. As a result, if a replicator runs only a subset of the workflow, the corresponding simulated outcomes remain reproducible.

To be precise, random seeds are set in the replication package at the following points:

- at lines 242, 308, 444, 587, 616, 924, 997, and 1461 of program `Master_model_work.jl`;
- at lines 109 and 547 of program `calib_functions.jl`.

3.3 Memory, Runtime, Storage Requirements

Hardware The replication package was last executed on a Mac mini with a 3.2 GHz 6-Core Intel Core i7 processor and 64 GB of RAM (2667 MHz DDR4).²

Runtime The total approximate time needed to reproduce the analyses on the computer referenced above is about 22 hours. More specifically, this is split as follows:

- Empirical analysis – approximately 12.5 hours, with the majority (about 10 hours) spent on the distance-to-default computations executed by `DD_work_par_localwrapper.py`;
- Structural model solution and analysis – approximately 9 hours, all spent on the model computations and analysis executed by `Master_model_work.jl`.

Storage Requirements The raw replication package, including the proprietary datasets, is approximately 11.5 GB in size. During execution, however, the empirical workflow creates updated, cleaned, and split copies of the CRSP data to facilitate parallel execution, while the structural model impulse response computations save extensive intermediate outputs and checkpoint files to reduce RAM pressure and allow the execution to be resumed with minimal loss of progress whenever interrupted. As a result, after the package has been fully executed, the combined size of the raw and generated files is approximately **120 GB**.

²Users running the replication package on Apple Silicon Macs (M1/M2/M3) who encounter setup difficulties may find the notes in Appendix A helpful.

Important Details The empirical workflow entails manipulation of the large daily CRSP dataset. The structural model impulse response computations at times store in memory individual firm *solution paths* over the idiosyncratic state space. For this reason, successful execution of the replication package requires a substantial amount of RAM. As noted above, the package was successfully executed on a machine with **64 GB of RAM**. In addition, the distance-to-default computations executed by `DD_work_par_localwrapper.py` are set up to run locally in parallel, with a maximum of 5 workers (set on line 16 of `DD_work_par_localwrapper.py`), so as to exploit the machine’s multi-core architecture. On machines with fewer cores, this step may run more slowly and may require reducing the number of workers.

4 Description of Programs/Code

- The program `Master_main.py` is the main driver file which executes the whole replication package. The majority of the work is organized in two branches:
 - the first, for the data cleaning and empirical work (executed by calling `Master_empirics_work.py`);
 - the second, for the computation and analysis of the structural model (executed by calling `Master_model_work.jl`).
- Once the analysis of the above branches is complete, `Master_main.py` calls a few remaining subprograms (detailed below) which require the presence of model-generated results.

4.1 Empirical analysis programs

This subsection lists the programs called in the empirical work of the replication workflow and briefly describes their respective roles. It includes both the programs called by `Master_empirics_work.py` in the main empirical branch and by `Master_main.py` after solving and analyzing the structural model.

- `code/Python/Master_empirics_work.py` – orchestrates the core empirical workflow, from data cleaning subprocedures through the firm-level local projection analysis.
- `code/Python/cleaning/CS_cleaning_code.py` – cleans the raw Compustat quarterly data and produces the preliminary cleaned Compustat dataset `CS_data_clean.csv`.
- `code/Python/cleaning/CRSP_cleaning_code.py` – cleans the raw daily CRSP data by dropping unused columns, and splits the dataset into smaller subfiles to reduce RAM pressure in later steps.
- `code/Python/cleaning/CStoCRSP_merge.py` – merges the cleaned Compustat balance-sheet data into the CRSP subfiles in preparation for computing distance-to-default.
- `code/Python/cleaning/CRSP_par_prep.py` – repartitions the CRSP-Compustat merged subfiles into a larger number of firm partitions to facilitate parallel execution of the distance-to-default computations.
- `code/Python/DD_work_par_localwrapper.py` – runs the distance-to-default computations in parallel over the repartitioned CRSP-Compustat subfiles by repeatedly calling `DD_work_run.py`.

- `code/Python/DD_work_run.py` – computes distance-to-default measures for a given partition of firms in the CRSP-Compustat merged daily panel.
- `code/Python/cleaning/Wscope_compile.code.py` – compiles the Worldscope incorporation and foundation date data into a single file used later to construct the firm-age control.
- `code/Python/Micro_reg_prep.code.py` – prepares the main empirical analysis panel by combining the cleaned Compustat data with the distance-to-default data, incorporation-date data, aggregate series, and monetary shock data.
- `code/R/Micro_reg_code.r` – conducts the main empirical analysis on the Compustat panel by constructing the final analysis variables and estimating the paper’s principal micro-level regressions and local projections. It also writes the resulting tables, figures, and supporting text outputs used in the manuscript and appendix, and exports the empirical event-study estimates and a subset of the local projection estimates.
- `code/Python/FOFA_analysis.code.py` – computes aggregate corporate liquidity measures from the Federal Reserve Flow of Funds Accounts for Table A.5 and for the pre-1990 counterfactual model liquidity holdings evaluation.
- `code/Julia/Julia_empirics.code/BDS_analysis.code.jl` – processes the Business Dynamics Statistics data used to construct the entrant-size calibration target reported in Table B.1.
- `code/Python/TB3_analysis.code.py` – constructs the counterfactual economy targets used in the structural-model analysis reported in Appendix B.10.1.
- `code/R/Micro_model_reg_code.r` – once the model is solved, conducts the debt issuance prediction and event-study regressions on model-generated firm-level panel data and exports the resulting event-study coefficients for later comparison with the empirical estimates.
- `code/R/Micro_eventstudy_plottr.r` – combines the empirical and model-generated event-study coefficient estimates and produces the comparison figures reported in Appendix A.3.
- `code/Python/Generate_TabB2.py` – combines the untargeted moments stored in `_partof_TabB2_nums.untargmoms.txt` and `_partof_TabB2_nums.untargmoms_model.txt` into the L^AT_EX body for Table B.2.
- `code/MATLAB/LPagg_main.m` – conducts the aggregate local projection analysis reported in Appendix A.6 and combines the resulting empirical impulse responses with the corresponding model-generated impulse responses.
- `code/MATLAB/printpdf.m` – auxiliary MATLAB function used by `LPagg_main.m` to export figures to `.pdf` format, as included in Luetticke (2021a).

4.2 Structural model analysis programs

This subsection lists the Julia programs used in the structural model branch of the replication workflow and briefly describes their respective roles.

- `code/Julia/Julia_model_code/Master_model_work.jl` – is the main driver program for the structural model analysis. It calls the functions that solve the firm’s steady state problem in the baseline and alternative model parameterizations, compute aggregate outcomes in steady states, solve impulse response scenarios, simulate firm-level panel data from the model, estimate model-implied heterogeneous response objects, and produce the model-side figures, tables, and saved solution objects used as checkpoints in the workflow.

- `code/Julia/Julia_model_code/auxiliary_model_functions.jl` – defines the core auxiliary procedures used throughout the structural model computations, including solution methods, interpolation routines, simulation procedures, equilibrium aggregation, and the construction of model-based regression and impulse response objects.
- `code/Julia/Julia_model_code/calib_functions.jl` – defines procedures used in computing model calibration moments, steady-state model diagnostics, lifecycle behavior, and group-specific aggregated impulse responses.

4.3 Additional Implementation Notes

This subsection records implementation details that are useful for replication but are not described fully in the paper or its appendix.

4.3.1 Implementation of Empirical Analysis

- **Duplicate resolution.** To construct a unique firm-period panel, the cleaning scripts resolve the small number of duplicate observations in the Compustat and CCM data as follows. When duplicate firm-date or firm-quarter observations are present and one record has missing `datacqr`, that record is dropped and the more complete duplicate is retained. After this step, and in the raw CRSP data, any remaining literal duplicates are removed by keeping the first or last occurrence. This duplicate-resolution rule is used only to address rare reporting-calendar-related duplications in the raw source data.
- **Distance-to-default construction.** The distance-to-default control is constructed from merged CRSP-Compustat daily data using a standard KMV/Merton-style procedure. Market equity is measured from CRSP prices and shares outstanding, and the face value of debt is proxied by current liabilities plus one-half of long-term debt. The procedure is applied only to firm-specific daily return segments with at least 252 observations and without extended gaps; gaps longer than one week are treated as breaks in the usable time series. Asset values and asset volatility are then recovered iteratively from the Black-Scholes relationship, using a one-year horizon, and the resulting distance-to-default measure is used as a control variable in the main robustness analysis.

In addition, because the distance-to-default measure is computed using an iterative root-solving procedure, very small machine-dependent numerical differences may arise across computing environments. When the resulting distance-to-default series is used as a control variable in the local projection regressions, these differences can lead to output files that are not bitwise identical across machines, although the resulting estimates remain visually indistinguishable.

- **Firm age construction.** The firm age control variable is constructed as time since firm incorporation as reported in Worldscope data. If the firm’s incorporation date is reported with only a monthly precision, the code sets the middle of the corresponding month to be the inferred incorporation date. If the date is reported with only a yearly precision, the middle of the year is set as the inferred incorporation date. The Worldscope data is matched to Compustat via CUSIP.

4.3.2 Implementation of Structural Model Computation

The notes below describe implementation choices made in solving, simulating, and analyzing the structural model that are useful for replication but are not described fully in the paper or its appendix.

- **Solution of the firm’s problem.** The individual firm’s problem in the structural model is solved numerically using approximation, interpolation, and iterative fixed-point methods suited to a high-dimensional heterogeneous-firm environment. The firm’s state space is four-dimensional, (k, m, b, z) , discretized on finite grids for numerical solution, where b denotes a conveniently defined adjustment of the beginning-of-period debt-to-capital ratio relative to the notation used in the paper.

More specifically, the solution procedure takes as given the current prices faced by the firm and interpolants for the expected continuation value function and its derivatives. The problem is then split into the two branches conditional on adjusting debt (A) or not (N), and the multivariate portfolio choice in each branch is decomposed into staged subproblems within the period. The firm’s optimal choices are then found either by explicit root-finding on the relevant first order necessary conditions or by inverting first order conditions using a multidimensional endogenous grid method (EGM) approach, extended with exogenous grids for Lagrange multipliers on financial constraints, analogously to the computational appendix of Auclert et al. (2021). To address potential non-convexities arising from debt issuance costs, the root-finding and EGM steps explicitly search over candidate solutions for the global optimum, following Fella (2014).

This same solution method is used both to compute steady-state firm policies and to solve firm behavior by backward iteration along impulse response paths, given the relevant current prices and expected continuation value objects. The steady state firm solution is obtained by starting from a guess for the value function and its derivatives, informed by the value function of an exiting firm, and iterating on these objects until convergence.

- **Interpolation.** Objects such as the firm’s expected continuation value function and its derivatives, as well as the firm’s policy functions, are computed on the discretized state space and linearly interpolated whenever they need to be evaluated off the grid.
- **Approximation of firm distribution.** The distribution of firms over the state space is approximated using the histogram method by Young (2010). To speed up the computation of the transition kernel of firms across this four-dimensional discretized space, I employ the algorithm proposed by Tan (2020).
- **Computation of general equilibrium responses to aggregate shocks.** To compute perfect-foresight general equilibrium responses to an unexpected aggregate shock realized at the steady state, the code uses a backward-solver / forward-iterator approach. Assuming that by some finite horizon T_g the economy is sufficiently close to steady state for truncation to be immaterial for the reported results, the code solves for price paths and aggregate quantities that satisfy equilibrium conditions at every $t = 0, \dots, T_g - 1$, under the terminal condition that from T_g onward prices and firm value functions are back at their steady-state values and remain there forever after.

More specifically, the model requires iteration on the paths of the capital price Q_t and the aggregate gross markup $\mathcal{M}_t$, denoted in the code by `pQ_col` and `MU_col`, until the capital

good and final good markets clear. Given any candidate paths for `pQ.col` and `MU.col`, the code computes all remaining price paths implied by equilibrium conditions such as the Taylor rule, the New Keynesian Phillips Curve, and the household labor-supply condition. It then backward-solves the firm’s problem for $t = 0, \dots, T_g - 1$ given these price paths, iterates the firm distribution forward, and infers the implied paths Q'_t and $\mathcal{M}'_t$ consistent with market clearing. If the candidate and implied paths are sufficiently close,³ convergence is taken to have been achieved. Otherwise, the candidate price paths are updated toward Q'_t and $\mathcal{M}'_t$ using strong dampening to ensure numerically stable convergence.

Because solving for such an equilibrium from a naive initial guess (for example, constant price paths at their steady-state values) is extremely time-consuming, each perfect-foresight general equilibrium computation in `Master_model_work.jl` is initialized with price-path guesses that already satisfy market clearing to the required tolerance. Concretely, the program includes these converged numerical series for `pQ.in` and `MU.in` as hardcoded initial guesses whenever perfect-foresight general equilibrium price paths are computed. The code then takes these guesses as given, solves the firm problem, iterates the economy forward, and confirms that markets clear. Each such backward-forward run takes about 10 minutes, whereas the full convergence procedure from a naive initial guess can take around 12 hours. As there are 9 different scenarios for which the replication package must compute general equilibrium price paths, this allows to save several days of runtime.

As a robustness check on the truncation horizon, during the development of the model, the solver has been run with both $T_g = 50$ and $T_g = 60$, showing that the resulting impulse-response paths were economically very similar, with no substantive differences in the responses of interest. The replication package therefore uses $T_g = 60$ as the baseline horizon.

- **Saved intermediate solution objects.** The structural model workflow employs saved checkpoint objects for computationally intensive solution steps. Baseline solution objects are stored in `interim_output/model_interim_output/savedsolns/` and may be reloaded before subsequent continuation steps. This checkpointing strategy is used to reduce runtime in case the code gets interrupted and to standardize the starting point for later model computations.

5 Instructions for Replicators

Full end-to-end execution of the replication package from raw sources requires access to the proprietary datasets used in the empirical branch of the workflow, and the presence of the corresponding raw files in the `proprietary-data-not-for-publication/raw/` directory (see Section 2.6).

To execute the replication package, proceed as follows:

1. Ensure that the software environments and package versions listed in Section 3.1 are installed.
 - (a) For the Python components of the workflow, a `requirements.txt` file is provided in the root directory of the replication package. The required packages can be installed via the terminal command `python3 -m pip install -r requirements.txt`.

³In the baseline computations, the stopping criterion requires successive iterations of price paths to differ by less than 0.002% of the relevant price level, point-by-point, i.e., a maximum log deviation between successive iterations of $2 \cdot 10^{-5}$.

- (b) For the R components of the workflow, an `install_R_packages.r` file is provided in the root directory of the replication package. The required packages can be installed via the terminal command `Rscript install_R_packages.r`.
 - (c) For the Julia components of the workflow, an `install_Julia_packages.jl` file is provided in the root directory of the replication package. The required packages can be installed via the terminal command `julia install_Julia_packages.jl`.
2. Preserve the original directory structure of the replication package, as described in Section 8.
 3. Set the `matlab_bin` variable on line 37 of `Master_main.py` so that it points to the MATLAB binary on the local machine.
 4. Execute `Master_main.py` in Python from the root directory of the replication package.

6 List of Figures, Tables and Additional Numeric Output

Table 2: List of Figures and Tables in the paper

Table / Figure	Program	Location in output/ folder
Main Text Figures and Tables		
Figure 1	<code>code/R/Micro_reg.code.r</code>	<code>figures/empirics_lumpiness/:</code> <code>Fig1a....pdf, Fig1b....pdf</code>
Figure 2	<code>code/R/Micro_eventstudy_plottr.r</code>	<code>figures/empirics_lumpiness/:</code> <code>Fig2a....pdf, Fig2b....pdf,</code> <code>Fig2c....pdf</code>
Figure 3	<code>code/R/Micro_reg.code.r</code>	<code>figures/empirics_LP/:</code> <code>Fig3a....pdf, Fig3b....pdf</code>
Figure 4	<code>code/Julia/Julia_model_code/</code> <code>Master_model_work.jl</code>	<code>figures/model_figures/:</code> <code>Fig4a....pdf, Fig4b....pdf</code>
Figure 5	<code>code/Julia/Julia_model_code/</code> <code>Master_model_work.jl</code>	<code>figures/model_figures/:</code> <code>Fig5....pdf</code>
Figure 6	<code>code/Julia/Julia_model_code/</code> <code>auxiliary_model_functions.jl</code>	<code>figures/model_figures/:</code> <code>Fig6....pdf</code>
Figure 7	<code>code/Julia/Julia_model_code/</code> <code>Master_model_work.jl</code>	<code>figures/model_figures/:</code> <code>Fig7a....pdf, Fig7b....pdf</code>
Figure 8	<code>code/Julia/Julia_model_code/</code> <code>Master_model_work.jl</code>	<code>figures/model_figures/:</code> <code>Fig8a....pdf, Fig8b....pdf</code>
Table 1	<code>code/R/Micro_reg.code.r</code>	<code>tables/Tab1....tex</code>
Table 2	<code>code/Julia/Julia_model_code/</code> <code>Master_model_work.jl</code>	<code>tables/Tab2....tex</code>

Table / Figure	Program	Location in output/ folder
Appendix Figures and Tables		
Figure A.1	code/R/Micro_reg_code.r	figures/empirics_lumpiness/: FigA1_...pdf
Figure A.2	code/R/Micro_eventstudy_plottr.r	figures/empirics_lumpiness/: FigA2a_...pdf, FigA2b_...pdf, FigA2c_...pdf
Figure A.3	code/R/Micro_eventstudy_plottr.r	figures/empirics_lumpiness/: FigA3a_...pdf, FigA3b_...pdf, FigA3c_...pdf
Figure A.4	code/R/Micro_eventstudy_plottr.r	figures/empirics_lumpiness/: FigA4a_...pdf, FigA4b_...pdf, FigA4c_...pdf
Figure A.5	code/R/Micro_reg_code.r	figures/empirics_LP/: FigA5a_...pdf, FigA5b_...pdf
Figure A.6	code/R/Micro_reg_code.r	figures/empirics_LP/: FigA6a_...pdf, FigA6b_...pdf, FigA6c_...pdf, FigA6d_...pdf
Figure A.7	code/R/Micro_reg_code.r	figures/empirics_LP/: FigA7a_...pdf, FigA7b_...pdf
Figure A.8	code/R/Micro_reg_code.r	figures/empirics_LP/: FigA8a_...pdf, FigA8b_...pdf
Figure A.9	code/R/Micro_reg_code.r	figures/empirics_LP/: FigA9a_...pdf, FigA9b_...pdf
Figure A.10	code/R/Micro_reg_code.r	figures/empirics_LP/: FigA10a_...pdf, FigA10b_...pdf
Figure A.11	code/R/Micro_reg_code.r	figures/empirics_LP/: FigA11a_...pdf, FigA11b_...pdf
Figure A.12	code/R/Micro_reg_code.r	figures/empirics_LP/: FigA12a_...pdf, FigA12b_...pdf
Figure A.13	code/R/Micro_reg_code.r	figures/empirics_LP/: FigA13a_...pdf, FigA13b_...pdf
Figure A.14	code/R/Micro_reg_code.r	figures/empirics_LP/: FigA14a_...pdf, FigA14b_...pdf

Table / Figure	Program	Location in output/ folder
Figure A.15	code/MATLAB/LPagg_main.m	figures/empirics_LPagg/: FigA15a....eps, FigA15b....eps, FigA15c....eps, FigA15d....eps
Figure B.1	code/Julia/Julia_model_code/ calib_functions.jl	figures/model_figures/: FigB1....pdf
Figure B.2	code/Julia/Julia_model_code/ auxiliary_model_functions.jl	figures/model_figures/: FigB2....pdf
Figure B.3	code/Julia/Julia_model_code/ auxiliary_model_functions.jl	figures/model_figures/: FigB3a....pdf, FigB3b....pdf
Figure B.4	code/Julia/Julia_model_code/ auxiliary_model_functions.jl	figures/model_figures/: FigB4....pdf
Figure B.5	code/Julia/Julia_model_code/ Master_model_work.jl	figures/model_figures/: FigB5....pdf
Figure B.6	code/Julia/Julia_model_code/ Master_model_work.jl	figures/model_figures/: FigB6....pdf
Figure B.7	code/Julia/Julia_model_code/ Master_model_work.jl	figures/model_figures/: FigB7....pdf
Figure B.8	code/Julia/Julia_model_code/ Master_model_work.jl	figures/model_figures/: FigB8....pdf
Figure B.9	code/Julia/Julia_model_code/ Master_model_work.jl	figures/model_figures/: FigB9....pdf
Figure B.10	code/Julia/Julia_model_code/ Master_model_work.jl	figures/model_figures/: FigB10....pdf
Figure B.11	code/Julia/Julia_model_code/ auxiliary_model_functions.jl, code/Julia/Julia_model_code/ Master_model_work.jl	figures/model_figures/: FigB11a....pdf, FigB11b....pdf
Figure B.12	code/Julia/Julia_model_code/ Master_model_work.jl	figures/model_figures/: FigB12....pdf
Figure B.13	code/Julia/Julia_model_code/ auxiliary_model_functions.jl, code/Julia/Julia_model_code/ Master_model_work.jl	figures/model_figures/: FigB13a....pdf, FigB13b....pdf
Figure B.14	code/Julia/Julia_model_code/ Master_model_work.jl	figures/model_figures/: FigB14a....pdf, FigB14b....pdf

Table / Figure	Program	Location in output/ folder
Table A.1	code/R/Micro_reg.code.r	tables/TabA1....tex
Table A.2	code/R/Micro_reg.code.r	tables/TabA2....tex
Table A.3	code/R/Micro_reg.code.r	tables/TabA3....tex
Table A.4	code/R/Micro_reg.code.r	tables/TabA4....tex
Table A.5	code/Python/FOFA_analysis.code.py	tables/TabA5....tex
Table B.1	code/R/Micro_reg.code.r, code/Julia/Julia_empirics.code/ BDS_analysis.code.jl, code/Julia/Julia_model.code/ Master_model_work.jl	Compiled manually, with inputs from: tables/: _partof_TabB1_BDS....txt, _partof_TabB1_CS....txt, _partof_TabB1_model....txt
Table B.2	code/Python/Generate_TabB2.py	tables/TabB2.body.tex
Table B.3	code/R/Micro_model_reg.code.r	tables/TabB3....tex
Table B.4	code/R/Micro_model_reg.code.r	tables/TabB4....tex
Table B.5	code/Julia/Julia_model.code/ calib_functions.jl	Compiled manually, with input from: tables/_partof_TabB5....txt

Table 3: List of additional numeric output supporting statements in the manuscript

Position	Program	Filename in output/other/ folder
Statements in Main Text		
Footnote 3	code/R/Micro_model_reg.code.r	Footnote3_justification.txt, Footnote3_justification_2.txt
Footnote 15	code/R/Micro_model_reg.code.r	Footnote15_justification.txt
Footnote 46	code/Julia/Julia_model.code/ Master_model_work.jl	Footnote46_justification.txt
Section 4.2 (text)	code/Julia/Julia_model.code/ Master_model_work.jl	Text_Sec4p2.txt
Section 5.1 (text)	code/Julia/Julia_model.code/ Master_model_work.jl	Text_Sec5p1.txt
Section 5.2 (text)	code/Julia/Julia_model.code/ Master_model_work.jl	Text_Sec5p2.GEPEshare.txt, Text_Sec5p2.rmrb.peak.txt, Text_Sec5p2.rmshresps.txt
Footnote 57	code/Julia/Julia_model.code/ Master_model_work.jl	Footnote57_justification.txt

Position	Program	Filename in output/other/ folder
Statements in Appendix		
Appendix (text)	A.2 code/R/Micro_model_reg_code.r	Text_AppxA2.txt
Footnote 61	code/R/Micro_model_reg_code.r	Footnote61_justification.txt
Footnote 74	code/R/Micro_model_reg_code.r	Footnote74_justification.txt
Appendix (text)	B.5.3 code/Julia/Julia_model_code/ Master_model_work.jl	Text_AppxB5p3.txt
Footnote 87	code/Julia/Julia_model_code/ calib_functions.jl	Footnote87_justification.txt
Appendix (text)	B.10.1 code/Python/F0FA_analysis_code.py, code/Python/TB3_analysis_code.py	Text_AppxB10p1_liqshares.txt, Text_AppxB10p1_rates.txt
Appendix (text)	B.11.1 code/Julia/Julia_model_code/ Master_model_work.jl	Text_AppxB11p1.txt
Appendix (text)	B.11.2 code/Julia/Julia_model_code/ Master_model_work.jl	Text_AppxB11p2_GEPEshare.txt

7 License

The code in this replication package is distributed under the MIT License. Data availability and redistribution remain subject to the restrictions described in Section 2.

8 Directory Structure

All scripts rely exclusively on relative paths and presume the directory structure shown below. The subfolders marked with “**” are created by the codes in the repository themselves.

```

.
|-- README.pdf
|-- LICENSE.txt
|-- Master_main.py
|-- requirements.txt
|-- install_Julia_packages.jl
|-- install_R_packages.r
|-- code/
|   |-- Julia/
|   |   |-- Julia_empirics_code/
|   |   |-- Julia_model_code/
|   |-- MATLAB/

```

```

| |-- Python/
| | |-- cleaning/
| |-- R/
|-- data/
| |-- raw/
|-- proprietary-data-not-for-publication/
| |-- raw/
| |-- processed**/
|-- interim_output**/
| |-- _DD_par_logs**/
| |-- model_interim_output**/
| | |-- savedout**/
| | |-- savedsolns**/
|-- output**/
| |-- figures**/
| | |-- empirics_LP**/
| | |-- empirics_LPagg**/
| | |-- empirics_lumpiness**/
| | |-- model_figures**/
| |-- other**/
| |-- tables**/

```

9 Key Variable Codebook

This section provides a compact codebook for selected constructed variables used in the main empirical and model analysis. It is not intended to reproduce full provider-level metadata for the underlying raw datasets, whose original variable documentation remains with the respective data sources. Instead, it documents the main derived variables that are central to the analysis and whose meaning may not be immediately obvious from source naming conventions alone.

Table 4: Key constructed variables used in the replication package

Variable	Dataset	Description / coding
<code>cheat_rat</code>	<code>Micro_reg_data_compiled.csv</code>	Cash-to-assets ratio, i.e. liquidity ratio. Constructed from Compustat cash and short-term investments (<code>cheq</code>) divided by total assets (<code>atq</code>).
<code>lev_rat</code>	<code>Micro_reg_data_compiled.csv</code>	Debt-to-assets ratio, i.e. leverage ratio. Constructed from Compustat total debt (<code>dltttq+dltcq</code>) divided by total assets (<code>atq</code>).
<code>k_stock</code>	<code>Micro_reg_data_compiled.csv</code>	Firm capital stock measure used in the investment and local projection analysis, constructed via perpetual inventory method (see Appendix A.1.2).

Variable	Dataset	Description / coding
diss_ind	Micro_reg_data_compiled.csv	Binary indicator equal to one if the firm issues long-term debt in the given quarter, and zero otherwise.
ndltisqat_rat	Micro_reg_data_compiled.csv	Net long-term debt issuance (constructed from the year-to-date Compustat variable <code>dltisy</code>) scaled by one quarter lagged assets (<code>atq</code>). Used both as an outcome variable and as an event indicator in the debt issuance analysis.
kinvat_rat	Micro_reg_data_compiled.csv	Quarterly net investment rate used for lumpiness and event-study analysis, constructed as change in <code>k_stock</code> relative to lagged <code>atq</code> .
invat_rat	Micro_reg_data_compiled.csv	Quarterly gross investment rate used for lumpiness and event-study analysis, measured as Capex (constructed from the year-to-date Compustat variable <code>capxy</code>) relative to lagged <code>atq</code> .
dcheat_rat	Micro_reg_data_compiled.csv	Quarterly change in cash and short-term investments divided by lagged assets, used for lumpiness and event-study analysis.
DD	Micro_reg_data_compiled.csv	Distance-to-default control variable constructed from merged CRSP-Compustat daily data using a KMV/Merton-style procedure.
age	Micro_reg_data_compiled.csv	Firm age, measured from Worldscope incorporation dates as time since incorporation.
sffr	Micro_reg_data_compiled.csv	Monetary policy shock series used in the firm-level and aggregate local projection analysis.
q_cheat_rat	Micro_reg_data_compiled.csv	Cross-sectional group indicator, based on terciles of cash-to-assets ratio distribution. Used in heterogeneous-response specifications.
q_Nlev_rat	Micro_reg_data_compiled.csv	Cross-sectional group indicator, based on terciles of net leverage ($(dlttq + dlcq - cheq) / atq$) distribution. Used in heterogeneous-response specifications.
cqtr_num	Micro_reg_data_compiled.csv	Numeric quarterly time index used as the main time variable in the empirical panel and in the construction of lags, leads, and event-time objects. E.g., 1995.25 refers to the end of 1995Q1.
gvkey	Micro_reg_data_compiled.csv	Compustat firm identifier used as the main firm-level panel identifier throughout the empirical analysis.

The model-generated panel dataset `df_model_panel.csv` is constructed so that its main variables mirror the empirical objects used in the firm-level model regressions and event-study comparisons.

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A Notes for Apple Silicon Mac Users (M1/M2/M3)

The replication package was developed and tested on an Intel Mac using the following language versions:

- Python 3.9.5
- R 4.1.0
- Julia 1.9.1

Most users should be able to reproduce the environment directly. This appendix documents a small number of setup issues that may arise on Apple Silicon Macs (M1/M2/M3), particularly when installing older software dependencies that were originally configured in an Intel-based environment.

A.1 Recommended Setup

On Apple Silicon Macs, the following configuration is recommended:

- Python 3.9.5, installed via `pyenv` or the official Python installer.
- Julia 1.9.1 using the macOS Apple Silicon (ARM64) build.
- R 4.1.0. If package installation fails under a native Apple Silicon build, users may wish to try the Intel (`x86_64`) build under Rosetta.
- GNU Fortran 8.2.0 for compiling R packages that require Fortran code.

A.2 Python

The Python code requires Python 3.9 or later. Some scripts use Python 3.9 language features and may fail under Python 3.8. Before installing Python packages, verify that the active Python version is 3.9.5. If using `pyenv`, a typical setup is:

```
pyenv install 3.9.5
pyenv local 3.9.5
```

```
python -m pip install --upgrade "pip<24" "setuptools<60" wheel
python -m pip install -r requirements.txt
```

If installation of `pandas==1.3.1` fails due to package build or dependency issues, the following commands may help:

```
python -m pip install "numpy==1.21.1" "Cython<3" \
    "setuptools<60" wheel
```

```
python -m pip install --no-build-isolation "pandas==1.3.1"
```

```
python -m pip install -r requirements.txt
```

A.3 R and lfe

Most users should be able to install the required R packages directly using `install_R_packages.r`; the following note applies only if package installation fails.

The R package `lfe` may need to be compiled from source. On some Apple Silicon Macs, installation can fail due to a missing Fortran toolchain. If installation fails with an error similar to

```
ld: library not found for -lgfortran
ERROR: compilation failed for package 'lfe'
```

then install GNU Fortran 8.2.0 (Intel/x86_64 version), and verify that

```
gfortran --version
```

returns:

```
GNU Fortran (GCC) 8.2.0
```

and rerun

```
Rscript install_R_packages.r
```

A.4 Julia

For Apple Silicon Macs, install Julia 1.9.1 using the macOS Apple Silicon (ARM64) build. Then run:

```
julia install_Julia_packages.jl
```

Some Julia packages may emit precompilation warnings. These are not necessarily fatal. If the installation completes and the replication scripts run successfully, such warnings can usually be ignored.

A.5 Final Verification

Before running the replication scripts, verify:

```
python --version
R --version
julia --version
```

The expected versions are:

```
Python 3.9.5
R 4.1.0
Julia 1.9.1
```

Then run the main replication script as described in Section 5.

Since the Python master script launches subprocesses using `sys.executable`, the master script should be started from the intended Python 3.9.5 environment.